

SOP-PT-4600: Hydrostatic and Pneumatic Pressure Testing

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Section 1. Purpose and Scope

Governs pressure testing of piping and vessels following new construction, repair, or alteration.

Section 2. Test Medium Preference

Clause 2.1 — Hydrostatic testing (water or other incompressible fluid) is the preferred method; pneumatic testing (compressed gas) is used only where hydrostatic testing is impractical (e.g., water intolerance, weight/support limitations) and requires additional written justification and engineering approval, given its materially higher stored-energy hazard.

Section 3. Test Pressure

Clause 3.1 — Test pressure is 1.5 times design pressure for hydrostatic tests (or per the applicable code for the specific piping class), held for a minimum of 10 minutes with no observed pressure drop, and visually inspected for leakage at all joints.

Clause 3.2 — Pneumatic test pressure shall not exceed 1.25 times design pressure per code, with a gradual pressure increase in staged holds, not a single rapid pressurization to test pressure.

Section 4. Exclusion Zone for Pneumatic Testing

Clause 4.1 — A pneumatic test requires a barricaded exclusion zone with no personnel present during pressurization above 50% of test pressure, calculated per the stored-energy hazard radius for the specific system volume and pressure.

Section 5. Records

Clause 5.1 — Test certificates (medium, pressure, hold time, result, witness signature) are filed to the Mechanical Integrity File and retained for the life of the equipment tested.